# Board Size Variation and Rates of Succession in the Corporate Presidency*

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*Within the setting of 54 industrial corporations, the higher the variation in the membership size of the board of directors, the more rapid the rate of replacement of the president. Variations in board size are developed from a coefficient of volatility identifying yearly changes from 1947 to 1977. Further analysis of the data suggests that for relatively successful companies a significantly negative relationship exists between board size variation and rates of succession, whereas a significantly positive relationship exists for relatively unsuccessful companies. These results have implications for succession planning in the corporate presidency.*

Pertinent to the study of the corporate leaders within complex organizations is the notion that viable systems seek self-stabilization while adjusting to the changing demands of the organization's task environment. In response to the changing environment, individuals are frequently hired and fired in organizational groups in order to introduce or make room for new administrative resources and ideas to deal with various contingencies. In this sense, organizations choose leaders who can help in dealing with critical contingencies that organizations face, suggesting that the choice of chief executives is systematically related to the changing dimensions of an organization's task environment. Pfeffer and Salancik [33], in summarizing Gouldner's [12] case study of succession, argued that "the frequency with which organizations replace their chief executive is hypothesized to be a function of the instability and problems confronted by organizations." Because organizations select leaders who can help in identifying critical contingencies of the environment [46], the choosing of a successor recreates the setting for examining the organization's mandate, effectiveness, and character.

Understanding the relationship between leadership changeover in key organizational groups and replacing leaders at the proper

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frequency is at the heart of effective succession planning, whether it be in the chief executive office or in the board membership. Periodic changes in leadership position are necessary for sustained organizational growth [2, 22, 23]. The exchange of people and information across boundary spanning groups of organizations is a significant means of ensuring management continuity and the link between organization and its environment.

**Hypothesis**

Pfeffer [31, 32] considered board size as one important organizational variable related to environmental contingencies confronting the organization because of varying requirements for environmental co-optation. Drawing from Selznick's [34] T.V.A. study, Pfeffer [31] observed that boards from time to time incorporate different facets of the task environment by adding or deleting directors who represent varying geographical, social, and economic sectors of the task environment. Additionally, Pfeffer [32] concluded that organizations are likely to be more successful in attracting resources to the extent that the board's composition is representative of the environmental context in which the organization is embedded. Vance [42] has labeled this use of directors with the desired access to environmental resources as "special economic service."

Zald [46] argued that the larger the number of members on the board having external power bases, the more likely were coalitions of board members to form—and the more coalitions, the more power. At the very least, according to Zald [46], the board will be involved in choosing the chief executive officer of the company when a replacement is necessary or desired. Significant to the current study is the contention by Zald [46] and Grusky [16] that at no time is the board's response to organizational policy change more visible than at the time of succession in the office of chief executive. Zald explained, "Since the mobilization of the board's influence occurs around the time of succession, the periodicity of succession becomes of greater importance" [46:109]. Since board membership varies in response to the changing organizational environment, since changes in corporate leaders are necessary to reflect those changes in the environment, and since boards do become involved in selection of successors, one may advance the following hypothesis: *variations in an organization's board size at the time of succession of the chief executive are related to the tenures of newly appointed successors.* The current research tests the implications by Zald [46] and Grusky [16] that successive changes in director membership parallel average rates of succession in the chief executive office. Succession rates are reciprocal measures of successive office tenures. Changes in both board size and chief executive tenure are considered to be responses in organizational adjustments to environmental conditions. Because of research [26, 31, 32] alluded to later in this inquiry, the advanced hypothesis is studied within the framework of both successful and unsuccessful companies along a profitability dimension.

When one considers that the board membership ebbs and flows over the history of the organization, the board becomes significantly vocal during crisis phases of development, especially when there is a need to choose a new successor. One might point out that the chief executive has a predisposition to nominate his or her successor ("crown prince"), but this is only done when the organization is in a stable state [46]. When an organization reaches a state of crisis due to environmental contingencies, nomination of a crown prince by the predecessor surely has the advanced consensus of the major stockholders on the board.

Besides the "crown prince" selection process, an additional category of variables directly influences chief executive selection. This consists of one of the varying forms of termination from office: death, illness, retirement, promotion, demotion, or dismissal. The last two reasons differ from the others in that they are measurable variables that are controlled by the organization. Promotion in the context of the current study is not a significant proportion of the sample base, especially since the sample is based on presidents who are chief executive officers. It would occur in the office of president, however, in the limited situation where a president of a subsidiary, for instance, moved into the presidency of the parent company. Retirement is partially controlled by the corporate organization in that some sort of succession planning has been implemented prior to the departure of the chief executive officer. Based on Newcomer's [29] classic survey, death and illness account for less than 30% of the terminations. These causes, then, are environmentally controlled, and they potentially introduce into a sampling procedure a disturbance in measuring aggregate succession rates. No prior reason is available to suggest that these two forms of termination are not randomly distributed over the companies surveyed in the present investigation. Adequate records for these particular forms of leaving office are difficult, if not impossible, to obtain for succession as far back as the 1930's.

## Methodology

**Sampling** Companies under analysis were selected from a listing of petrochemical companies engaged in the manufacture of petroleum derivatives, refining plastics, and chemicals and allied chemical products. This list is available in any issue of *Moody's Industrial Manual* under "Classification of Companies by Industry and Products" [28]; however, the most recent (1977) edition was used for this sampling aspect of the study. The petrochemical industry itself is interesting to examine for several additional reasons. (1) Productivity statistics [37] for the United States show that the annual percentage of gross national income accumulated by the chemical and petroleum industries for the past 10 years is growing faster in real value than that of most other industries. To a certain extent this evidence suggests that the petrochemical industry is adjusting more quickly to the demands of the environment than are many comparable economic sectors. (2) Except for the 1930's the chemical industry has earned a reputation as a growth leader [9]. (3) Early studies of Blauner [4], Udy [41], and Woodward [45] in aspects of technology, structure, and formalization, suggest that the continuous process nature of the petrochemical industry would tend to homogenize administrative process in terms of the organization's dealings with selection, retention, and replacement of executives.

Lawrence and Lorsch [26] have felt that the entire chemical industry is too diverse for sampling in terms of environmental forces upon varying size organizations. Therefore, some 300 companies were initially selected from those listed in *Moody's Industrial Manual* [28] – 1947 through 1977. Companies listed must have assets over one million dollars. Similar to those explanations provided in Lieberson and O'Connor's [27] *ex post facto* study, there are reasons for extending the time period of the study only back to 1947. (1) Given several studies [20, 23, 36] that find the average time in office for the chief executive officer to be approximately six years, it seems reasonable to provide on the average five or so successions per company in order to compute a mean succession rate. (2) The 1947 calendar year just follows the turbulent period (production restrictions, recruitment limitations, and so forth) of World War II. (3) The extension of the lower cutoff before 1947 would tend to elevate the probability of corporate reorganization, nonavailability of data, and variances in technology and product lines due to war-time technology. Deletions from the initial listing were made if a complete history under analysis was not included in *Moody's* editions from 1947 through 1977 due to vacancies in the presidency and lack of identification of chief executive officers. Finally, companies were not considered for analysis and subsequent data collection if merger activity radically changed the administrative structure or if bankruptcy took place during the succession periods used in the analysis. In light of these factors, a sample of 54 companies was identified for further study.

## Operational Definitions

The average succession rate for any given company was found by aggregating (averaging) the reciprocals of *completed* office tenures during the 30-year interval for all presidential successors who held the title *Chief Executive Officer*. Office tenure for successors was determined by examining selected editions of *Moody's Industrial Manual* [28]. The specification of the 30-year interval was also statistically necessary because of the need to control the time variable across companies when developing *ex post facto* measures of succession rate and variability in measures of board size. Operationally, board size simply included the number of directors serving on the board during the year of measurement. The data for board size variation as well as for average succession rates were developed from *Moody's Industrial Manual* [28] and *Standard and Poor's Register* [35].

Variations in the board of director size requires some further analysis. Viewing the footnoted equation[1] for the variation coefficient for board size, one can assign yearly values of board size to the variables $Y_i$ for the 30-year time period from 1947 to 1977.

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[1] The modified coefficient of variability for a given company in the sample is as follows:

$$c.v. = \frac{\sqrt{\dfrac{\Sigma(y_i - y_c)}{N-k}}}{\bar{y}}$$

$Y_i$ = variable measure of board size
$\bar{Y}$ = mean of $Y_i$'s
$Y_c$ = linear predicted value of $Y_i$ (least squares)
$N$ = number of $Y_i$'s
$k$ = number of $b$-coefficients in predictor model ($k$ = 2 for linear equation)

The prediction ($Y_c$) of $Y_i$ for each variable was accomplished by linearily regressing a particular variable under calendar time in years, starting from 1947 up to and including 1977. Yearly changes were considered because *Moody's Industrial Manual* [28] is only published annually. This is a slight modification from Tosi, Aldag, and Storey's [39] yearly measure first used to find volatility coefficients. Developing the residuals ($Y_c - Y_i$) not only accounts for annual variation in board size around a nominal size but corrects for random noise in the data and any long-term linear trends due to organizational growth or decline. (This study does not view growth or decline in board size by only variation due to environmental changes.) Moreover, residual indicators account for the fact that some companies have larger boards than others. The final form in which a mean residual standard deviation in the numerator is divided by the mean ($\bar{Y}$) of the $Y_i$ permits comparability across companies. Correction for $N - k$ degrees of freedom enables the variation measure to be used with relatively small sample sizes, with two degrees of freedom ($k = 2$) lost for the regression coefficient used in the development of a linear equation prediction ($Y_c$) of $Y_i$.

## Potentially Confounding Influences

In developing the hypothesis relating variations in size of the board of directors with average rates of succession, one should observe that prior research has shown that certain organizational variables confound the relationship under study. Although the operational definition of board size variation does account for differences in boards of directors in terms of size from company to company, prior studies [1, 8, 11, 13, 14, 16, 19–21, 25, 40] have demonstrated that large organizations find it more difficult to change the size of the board of directors and also tend to experience a greater rate of replacement in key corporate positions because of the necessity of succession planning. Similar studies [10, 15, 18, 22, 23, 27, 47] also present evidence to suggest the potential interaction of both organizational growth and performance upon the two major variables under study. Because some organizations are less willing to change their administrative structure with age [8, 43], it seems reasonable to partial out the potential effects of organizational age in viewing the hypothesized relationship.

A final potentially confounding variable to the particular administrative process under consideration in this inquiry relates to the configuration of an organization's technological core [4, 41, 45]. Because directors are usually added or removed from boards to adjust to changes in product lines [42], it seems necessary to partial out the effects of "product diffusion" by recording Standard Industrial Classification (SIC) indices for the companies under analysis. Realizing that some product lines are neither always identifiable nor reported to industrial directories, one may still categorize product differentiation by developing a range or standard deviation for listed SIC codes. The validity of this measure is seen in Berry's [3] 1974 study of corporate growth and product diversification, where SIC codes are analyzed and related to Herfindahl's "Index of Industrial Concentration" [24]. Further details of this variable, along with details of other confounding variables noted above, are presented in Table 1 below in conjunction with statistics on all variables considered to this point in the investigation.

## Results and Discussion

In considering Table 2, one observes a significant positive influence ($b = 0.254$, $F = 8.63$, $P < 0.05$) for board size variation upon the rate of succession in the office of chief executive over the 30-year interval of study for the total sample of 54 companies. That is, the higher the variation in board size, the more rapid the change in leadership in the chief executive office. This conclusion supports the advanced hypotheses based on the implications [31, 32, 34] that the board size variation, as a measure of adjusting the internal organization structure, is further linked with changes in leadership as one means of responding to demands of either the task environment of the internal organization itself. The argument that companies adjust both their board membership and frequency of replacing presidents who are CEO's is supported within the current sample base. This finding indicates that while the argument whether or not the board of directors is the main vehicle for organizational direction [30–32, 42, 46], variations in its membership size appear to have ramifications for office tenures of corporate presidents. Environmental changes in the chief executive office, and this reasoning seems to hold regardless of the size, growth, performance, age, and product diversification of the company.

Because of the significance of the relationship between board size variation and rates of succession, one can observe implications

TABLE 1: Descriptive Statistics for Variables Under Analysis ($N$ = 54)

| | Average Succession Rate | Board Size Variation[b] | Organizational Size[c] | Organizational Growth[d] | Organizational Performance[e] | Organizational Technology[f] |
|---|---|---|---|---|---|---|
| Variable Mean | 0.207[a] | 0.275 | 1046.7 | 164% | 1.5% | 433.7 |
| Variable Standard Deviation | 0.101 | 0.244 | 1485.7 | 214% | 5.9% | 478.8 |

[a] This mean is equivalent to an average time in office of 5 years (= 1/0.207).

[b] Mean and standard deviation for board size aggregated over interval 1947–1977 are 11.95 and 4.2, respectively.

[c] In millions of dollars of sales volume averaged over time interval of study.

[d] Percent increase in assets over time interval of study, using 1947 as base year.

[e] Average profitability (expressed as percent net income before taxes less extraordinary gains or losses over gross sales) during time interval of study.

[f] Average standard deviation (root-mean-square) of yearly standard deviations for Standard Industrial classification codes (S.I.C.) over time interval of study.

TABLE 2: Standard Partial b-Coefficients for Regression of Succession Rate on Board Size Variation

| | Total Sample ($N$ = 54) | | Dichotomized on Profit: Unsuccessful ($N$ = 27) | | Dichotomized on Profit: Successful ($N$ = 27) | |
|---|---|---|---|---|---|---|
| | b-coef. | F-test | b-coef. | F-test | b-coef. | F-test |
| Board Size Variation | 0.254[a] | 8.63** | 0.603[b] | 5.5** | −0.230[c] | 6.97** |

[a] All background variables measuring organizational size, growth, performance, and technology are partialled out simultaneously.

[b,c] All background variables are partialled out simultaneously except for organizational performance.

** $p < 0.05$.

for the process of succession planning in the corporate structure. However, also important for succession planning is the question of whether profitable companies exhibit this positive relationship or whether it is a phenomenon independent of corporate profitability. We have already cited Pfeffer's [31, 32] observations that organizations are more likely to be successful if the organization is flexible in exchanging information with the environment and with other organizations via changes in board composition and board size. Lawrence and Lorsch [26] had noted that successful organizations tend to have internal structural changes that are congruent with environmental demand and that any deviations from a certain pattern may affect performance. The sample of 54 companies is dichotomized along a profitability dimension as presented in the right-hand portion of Table 2. Advocating parallel changes in board size with concurrent changes in presidents would not make sense if this relationship were found to be a characteristic of unsuccessful rather than successful firms.

The profitability dichotomization of the sample ($N$ = 54) produces some useful results that, like other research [11, 21] in this field, may be of even greater interest than the initial hypothesized relationship. For example, among the relatively successful companies a significantly negative $b$-coefficient ($b = -0.23$, $F = 6.97$, $P < 0.05$) occurs when variations in board membership are regressed on the rate of succession. The higher the degree of variation in the board membership, the less the rate of succession (longer tenure), and vice versa. However, for relatively unsuccessful companies the finding ($b = 0.603$, $F = 5.5$, $P < 0.05$) suggests that high rates of succession follow relatively high degrees of variation in the board membership and vice versa.

These results suggest that, for succession planning, relatively high performing organizations either (1) seek out replacements of the chief executive officer at a faster rate if the board membership is fairly stable or (2) maintain current chief executives in office longer during periods of turbulence in the membership of the board of directors. These results for the successful companies tend to indicate that the best pattern of organizational change is to trade off shifts in membership in the board of directors with tenure of the chief executive officer. This strategy would be executed rather than (1) varying the membership of the board and frequently changing the chief executive officer or (2) restraining variations in the board membership and maintaining the infrequent variation of office of the chief executive. Some support for successful companies executing frequent change in the office of chief executive while maintaining stability in the board membership is supported by Grusky's [13, 14] work on succession. Here, the argument is proposed that efficient bureaucracies seek management change at the top because they can withstand leadership change due to rigidity in organizational structure. More important is the support for Pfeffer's [31, 32] finding that organizations which deviate from a preferred situation in their board of directors tend to be less significantly profitable. The preferred situation of the board is stability when changes are taking place in the office of the corporate president.

Another consideration possibly giving rise to the negative $b$-coefficient ($b = -0.23$, $F = 6.97$, $P < 0.05$) for successful companies is proposed by Ziller's [47] study of open and closed groups. He suggests that to the successful group, the newcomer is a resource to the current organizational leadership and therefore is encouraged to stay when change is desired. That is, the new leader is brought in to make changes in the board of directors. In this sense, longer chief executive tenure is a reflection of his ability to frequently change the membership size of the board. Because the company is operating in a successful mode, it appears that there is not immediate necessity of repeatedly replacing the chief executive officer. According to Zald [46], the only real association between the board membership and the organization is through the CEO, and one of the prime responsibilities of boards is the frequent replacement of the CEO. The amount of active participation in replacement of chief executive leaders is probably a good index of the board's power and stability. And the inference follows that relatively successful companies trade off chief executive tenure with board membership size, whereas unsuccessful companies tend to make parallel changes in both board membership size and rates of presidential replacement. What one sees in the current research is the notion that companies in general exhibit a relationship between board size and presidential succession rates, but that this relationship changes in direction when companies are classified into successful vs unsuccessful categories. In this sense what is characteristic of the industry is not necessarily characteristic of relatively successful companies, and certainly relatively unsuccessful organization might consider this aspect in planning for succession in the presidency.

## Conclusion

A positive relationship has been considered between variations in the size of a company's board of directors and the rate of replacement of the corporate president for those carrying the title of chief executive officer. The results support the positive relationship, but when the sampled companies were categorized into unsuccessful vs successful companies, the inferences are modified. The hypothesized relationship between board size variation and rates of presidential succession held for unsuccessful companies but not for successful companies. This suggests that in planning for presidential succession, either (1) successful organizations exhibit a degree of rigidity in board membership size, increasing the likelihood that successions will be routinely treated and routinized into frequent replacements of chief executive officers, or (2) instability through membership size change in the board is reflected in infrequent changes of chief executive officers.

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